Training one agent per time

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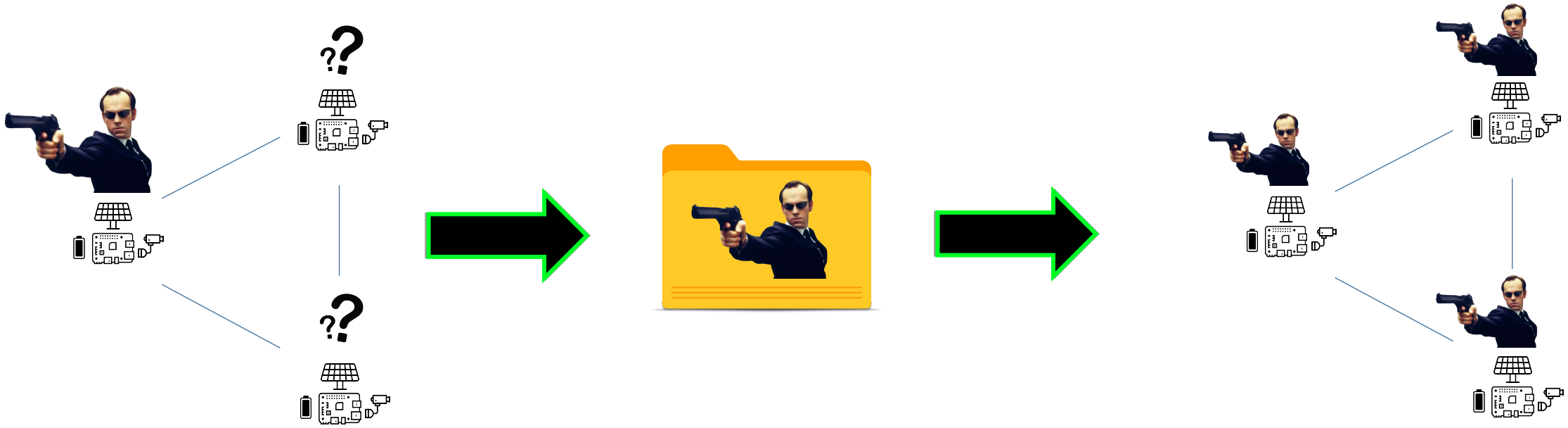
Key idea

- Given N different agents, let N-1 of them act randomly and the remaining one learns a policy
- Saving trained model and reusing for future training with different configurations
- Loading trained model in all the agents’ NN for evaluating obtainable performances

Key idea: a sketchy representation

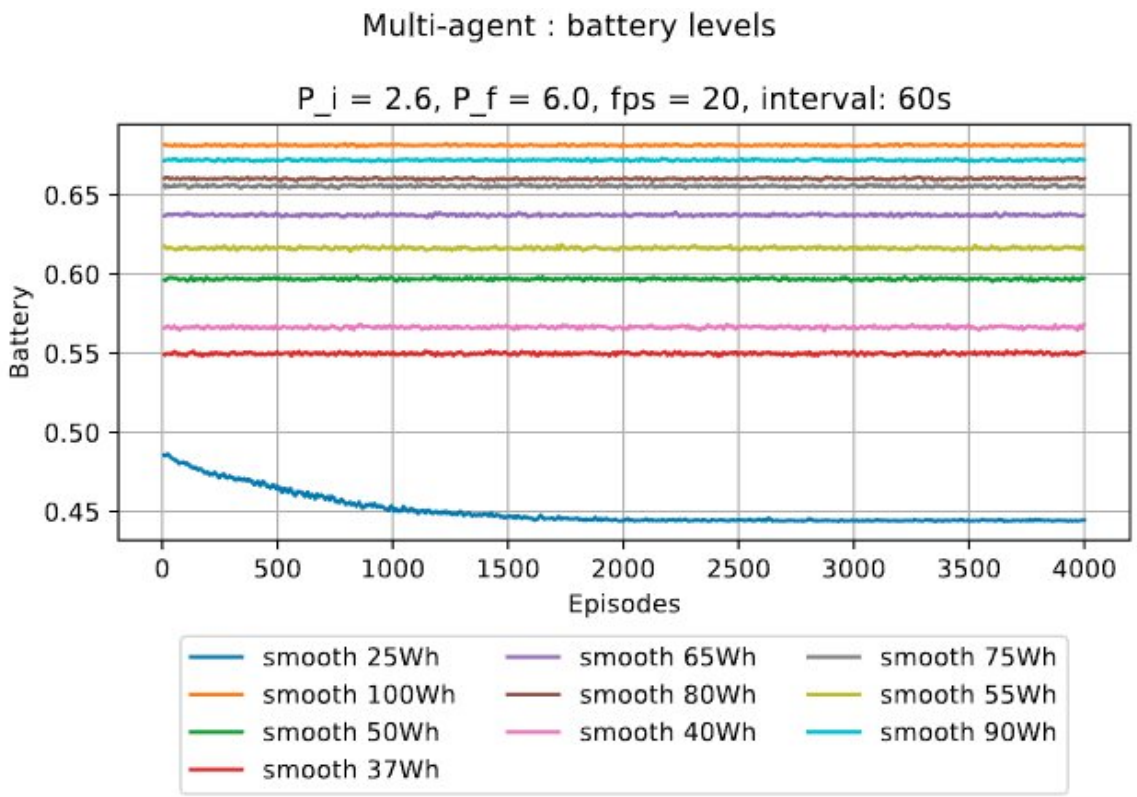
- Training only one agent, N-1 have not to learn
- Saving the model
- Loading the model for future trainings

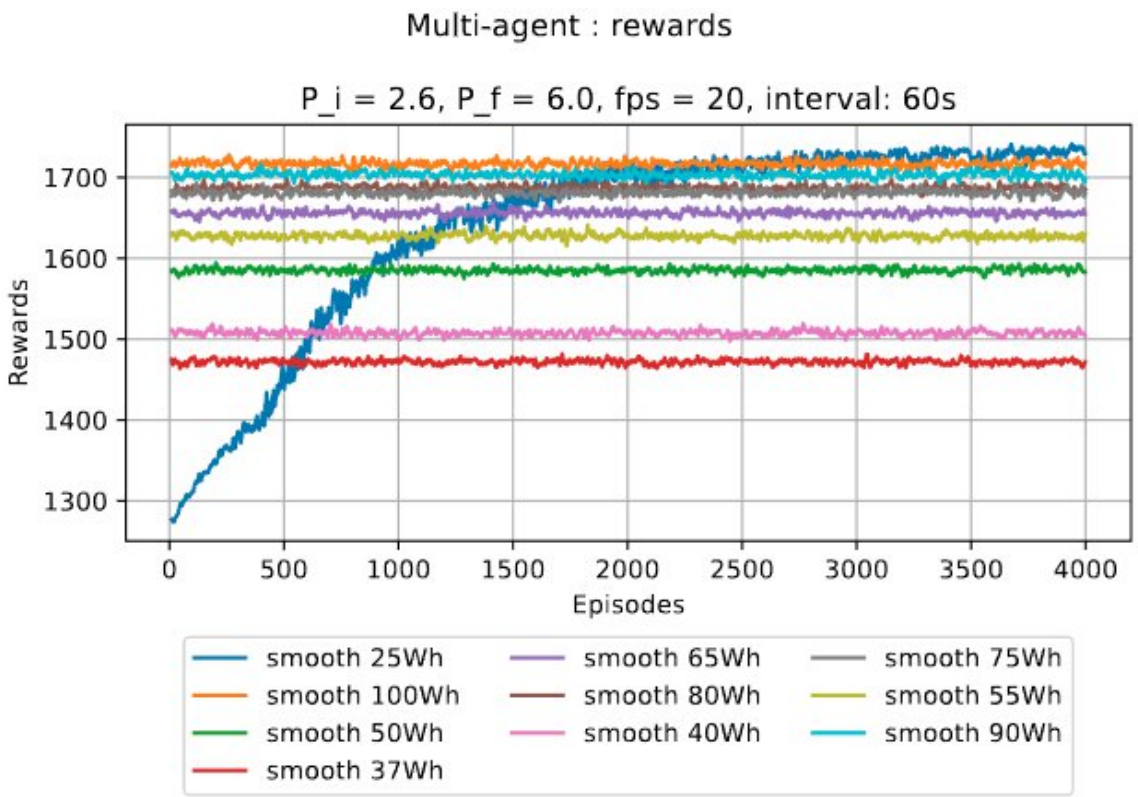


System setup

- N = 10 agents
- Training sessions run of 2000 episodes
- Battery capacities: [25, 100, 50, 37, 65, 80, 40, 75, 55, 90] Wh
- Panel surfaces: [1.00, 0.50, 0.75, 0.85, 0.65, 0.55, 0.90, 0.30, 0.80, 0.52] m²
- Observations: $o_t^i = [b_t^i, q_t^i, \tau_t^i, \mu b_t^i, \mu q_t^i]$
- Usual solar data from Solcast, 355th day of 2024
- Irradiance data with **noise of ±20%** wrt dataset values

Initial test: intelligent 25Wh





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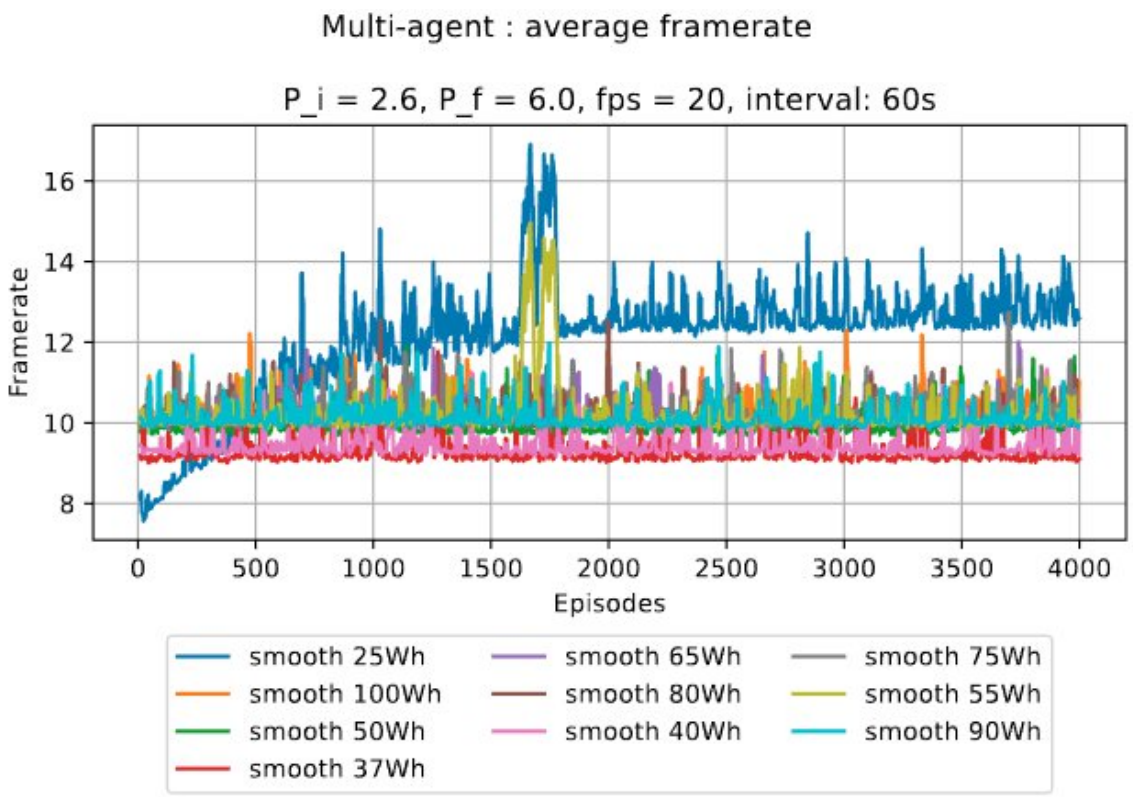
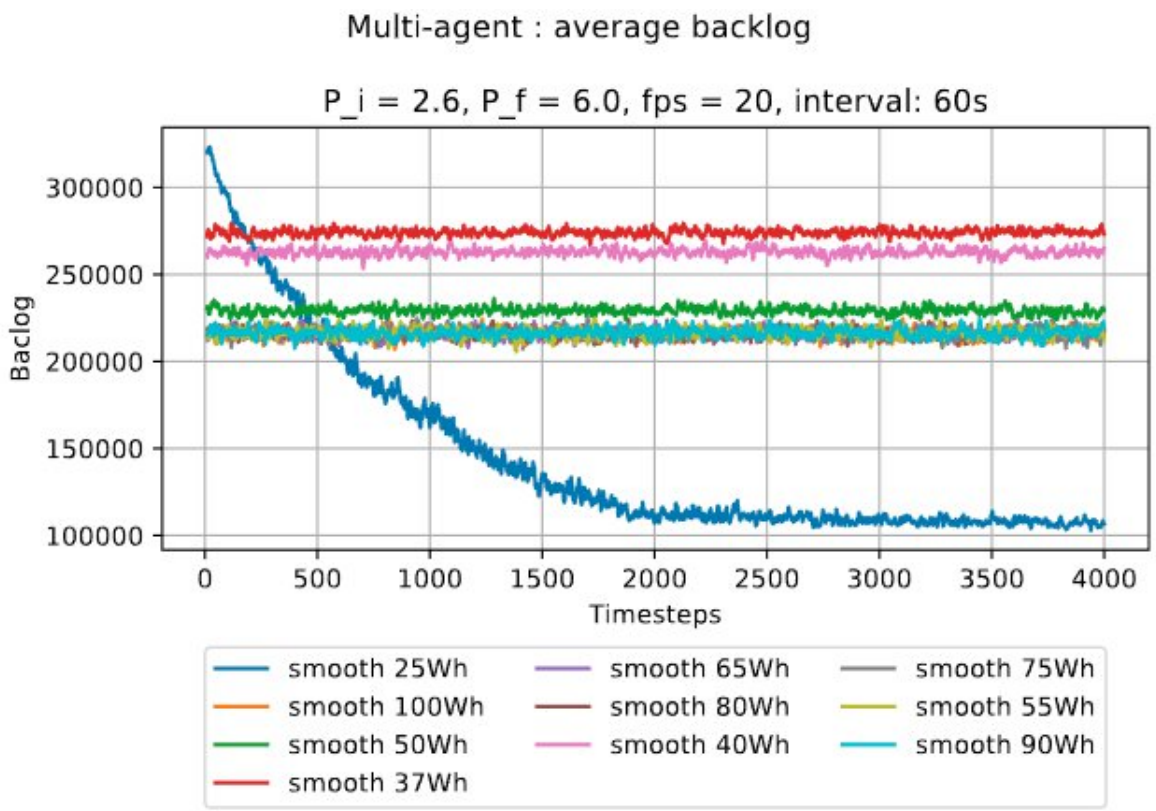
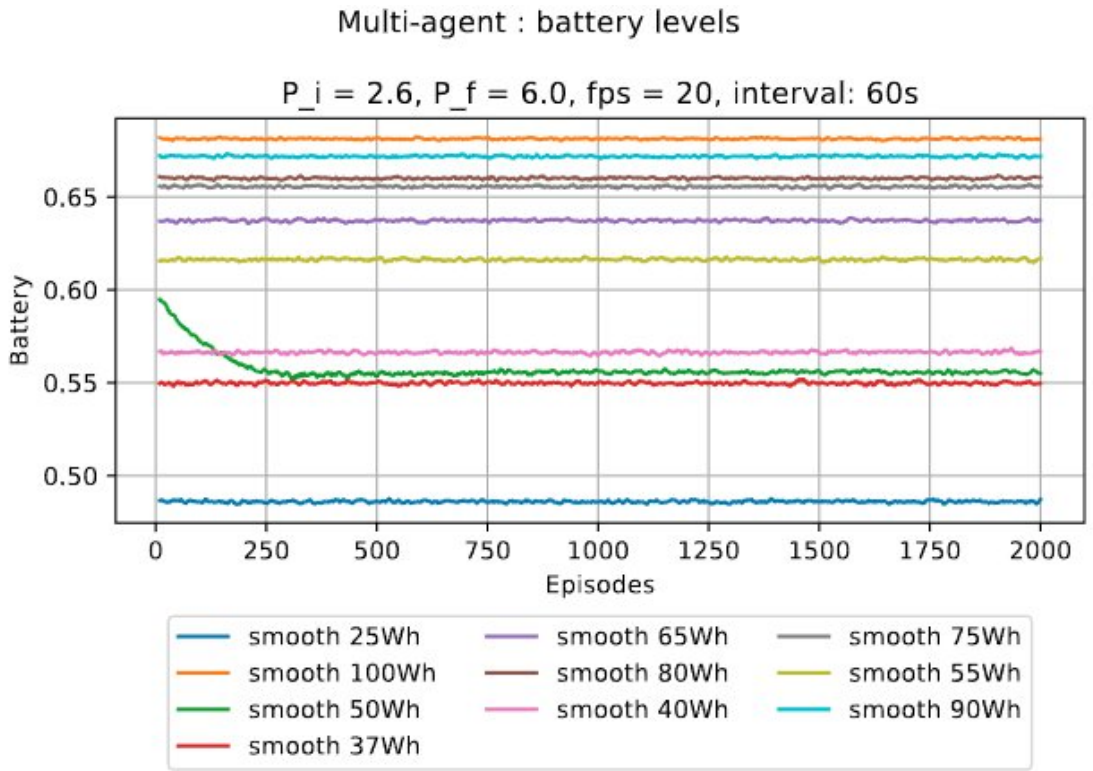
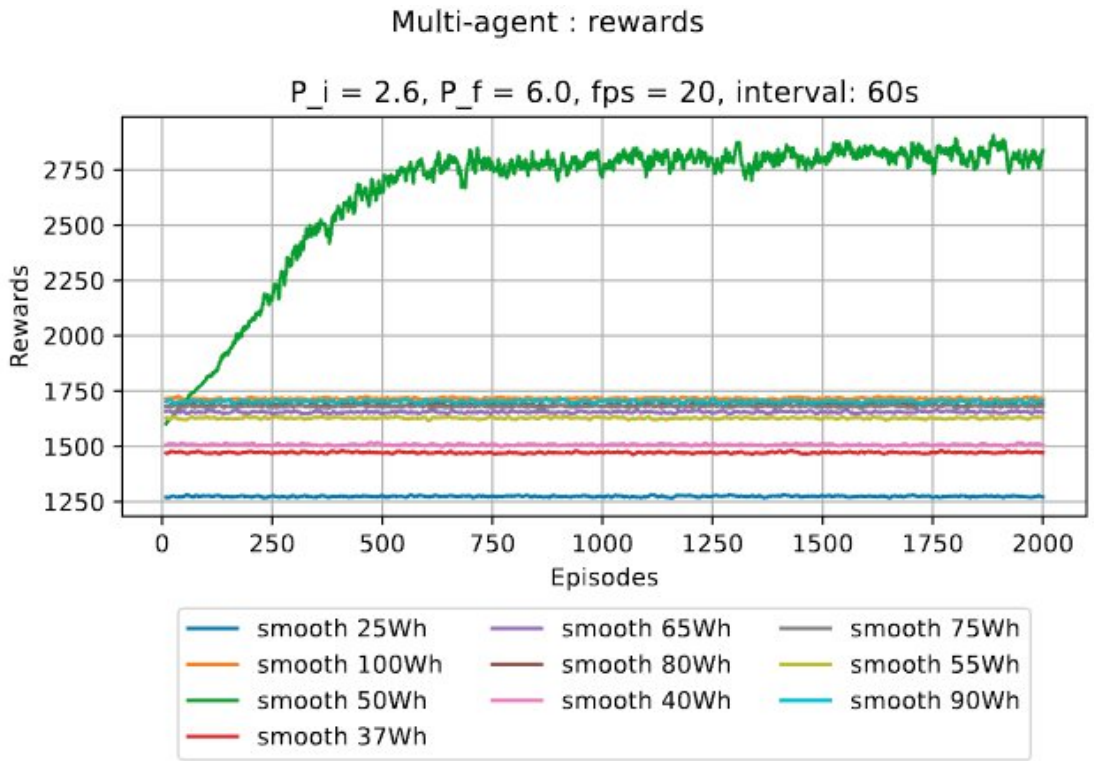




Second test: intelligent 50Wh

- Imported trained model of node with 25Wh





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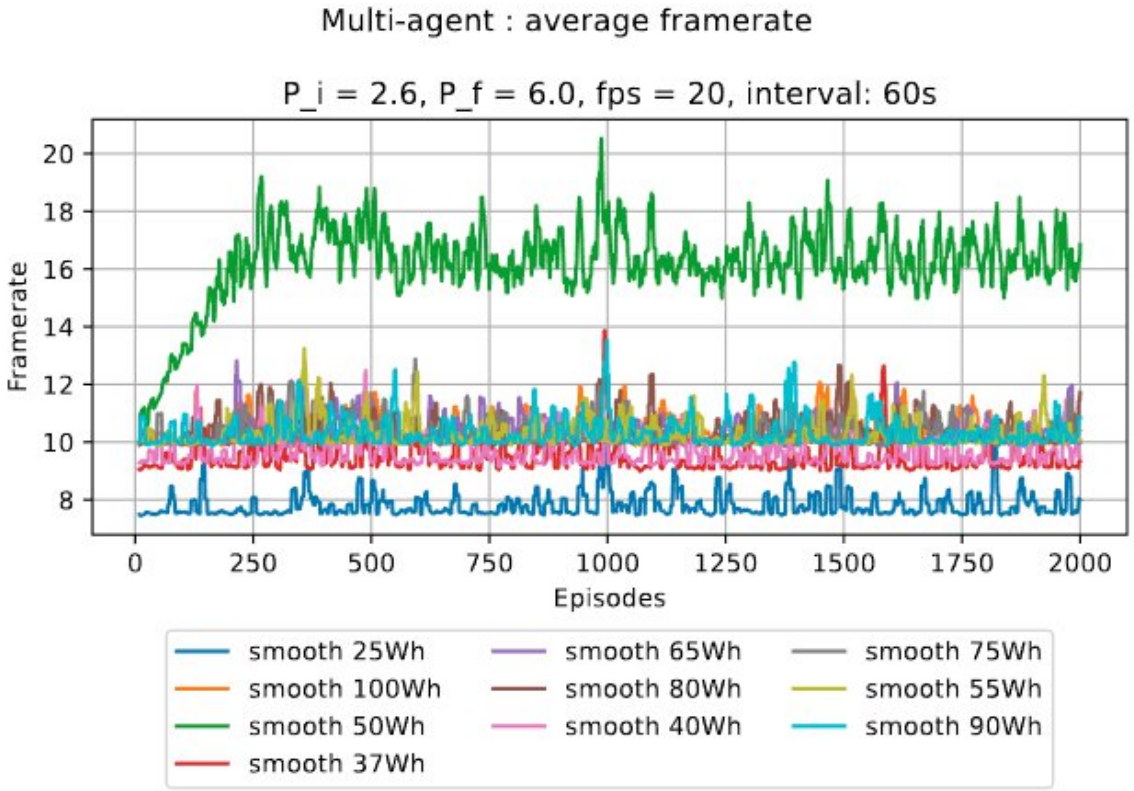
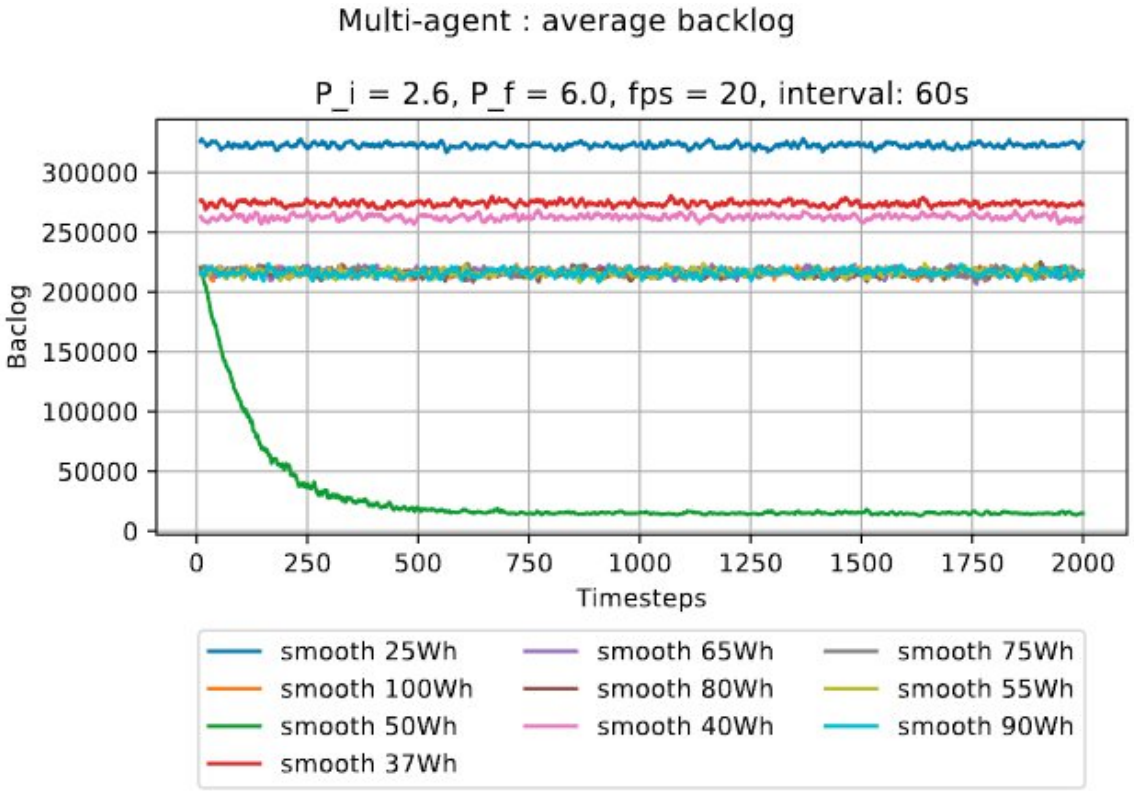
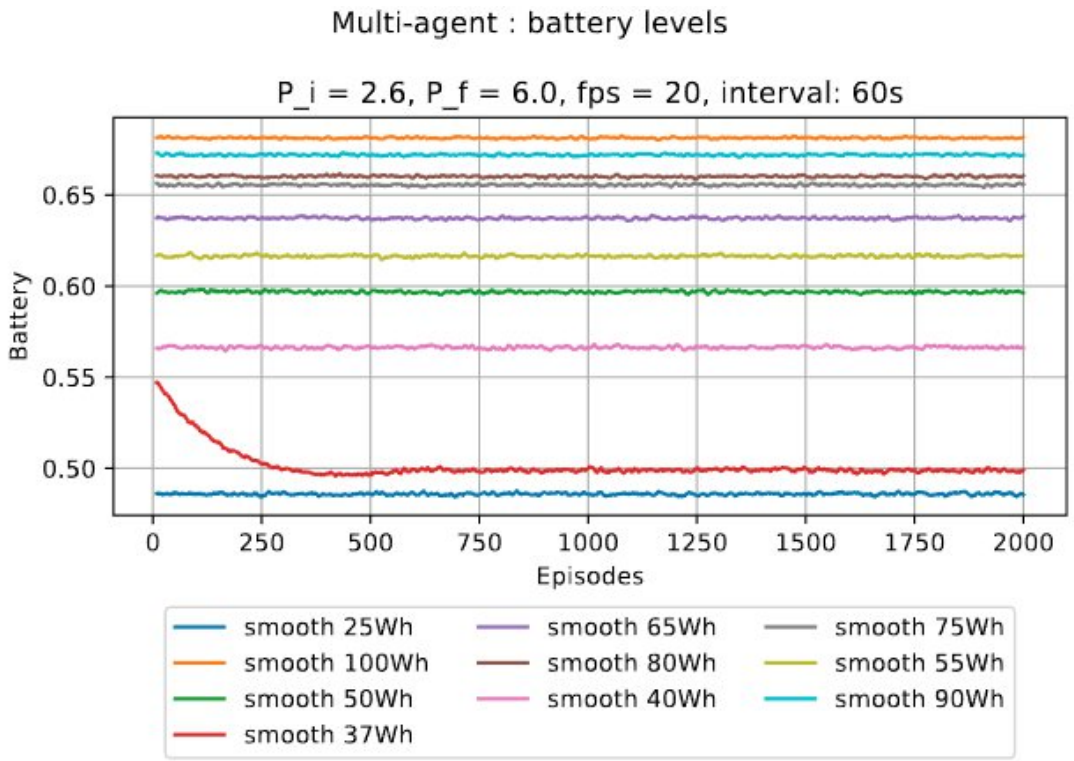
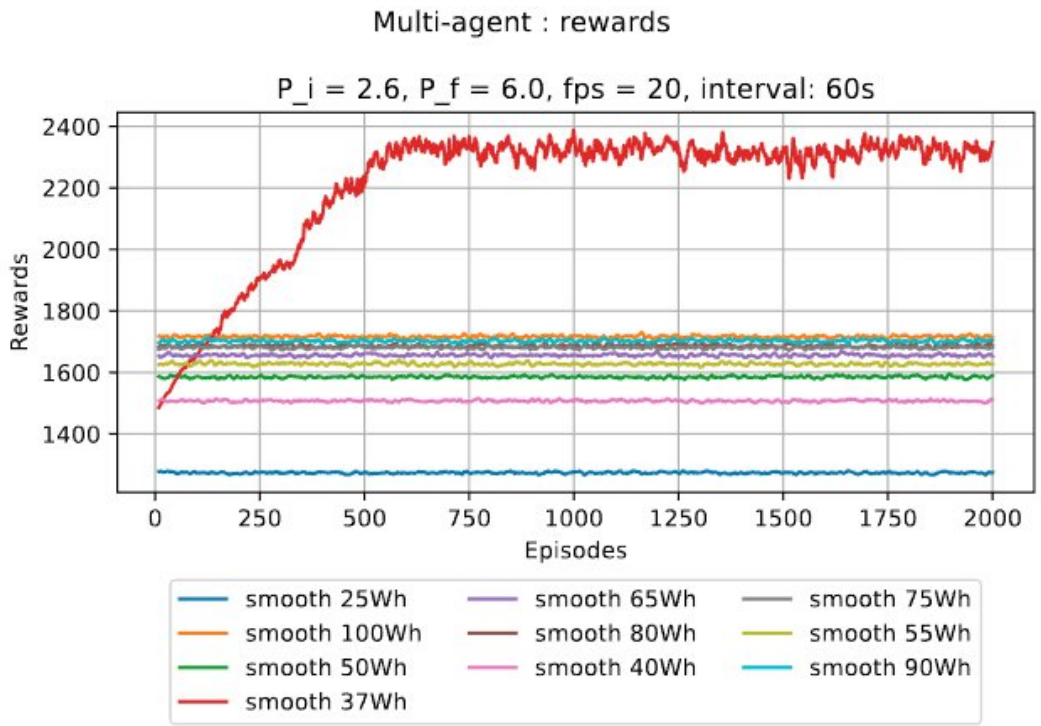




Third test: intelligent 37Wh

- Imported trained model of node with 50Wh





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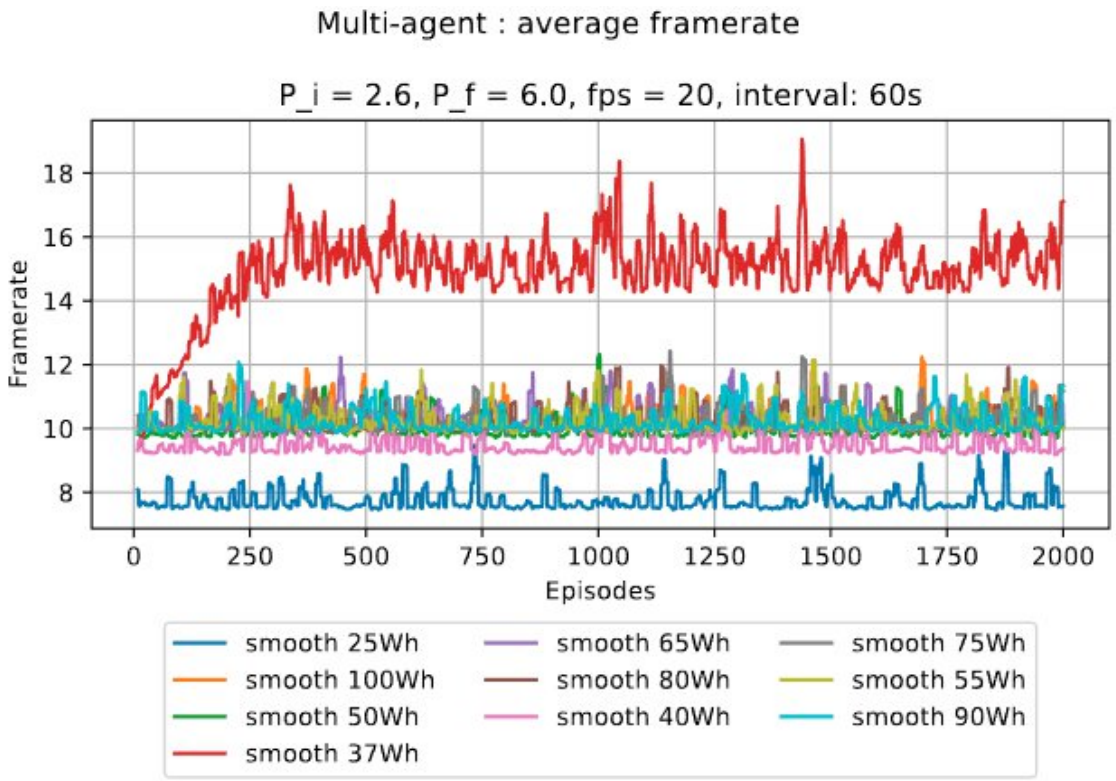
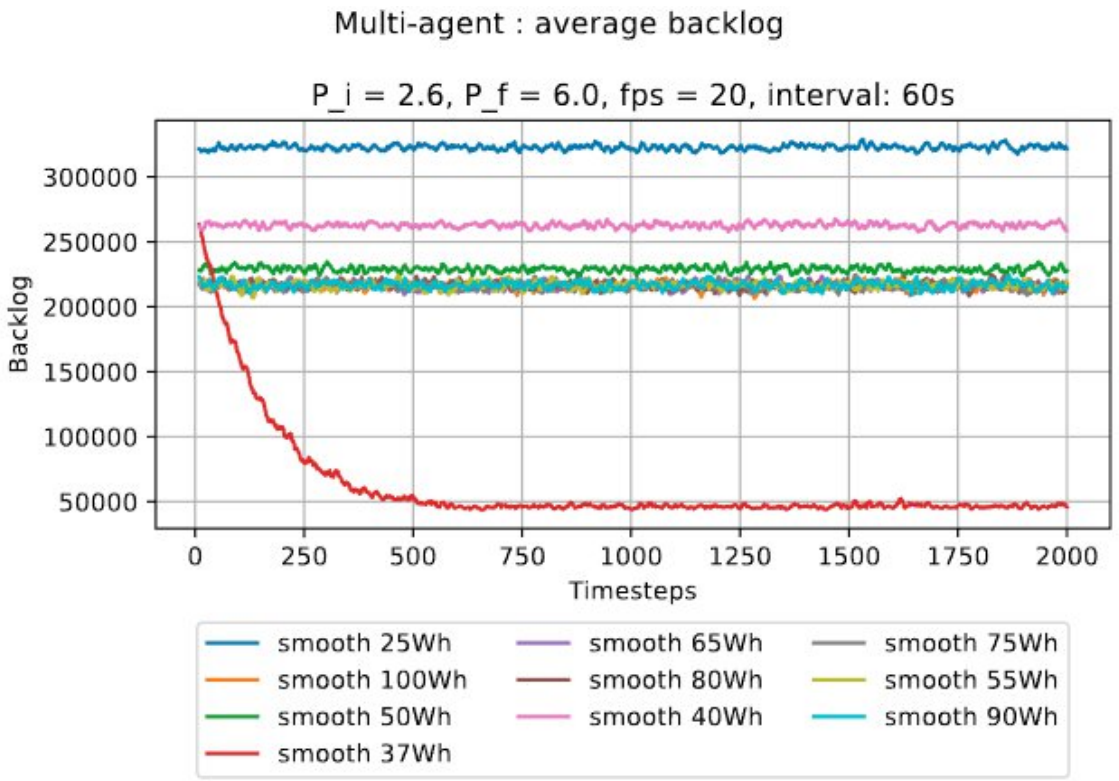
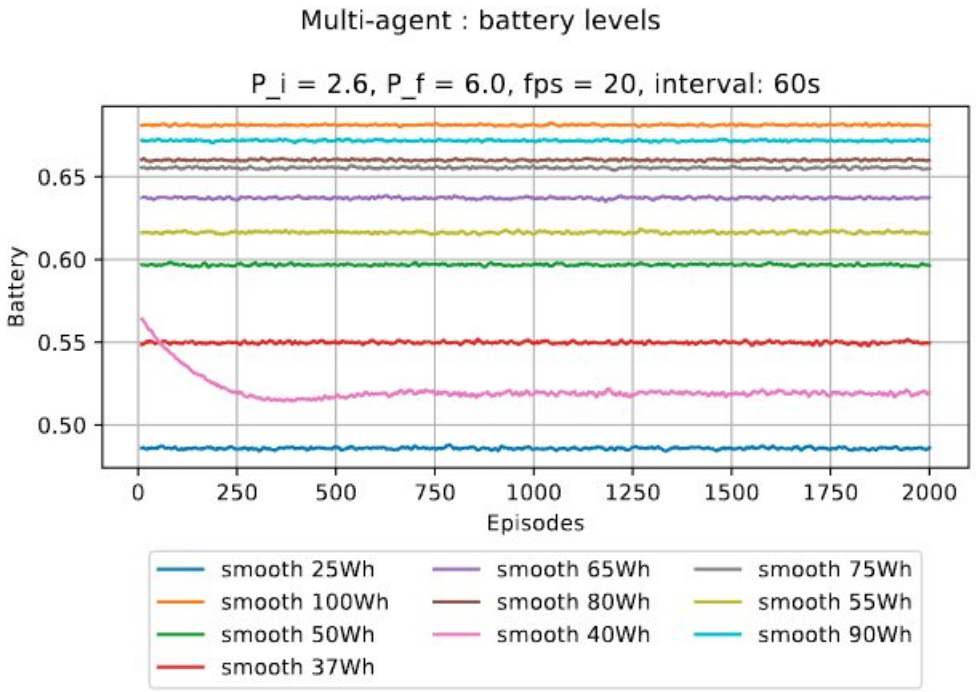
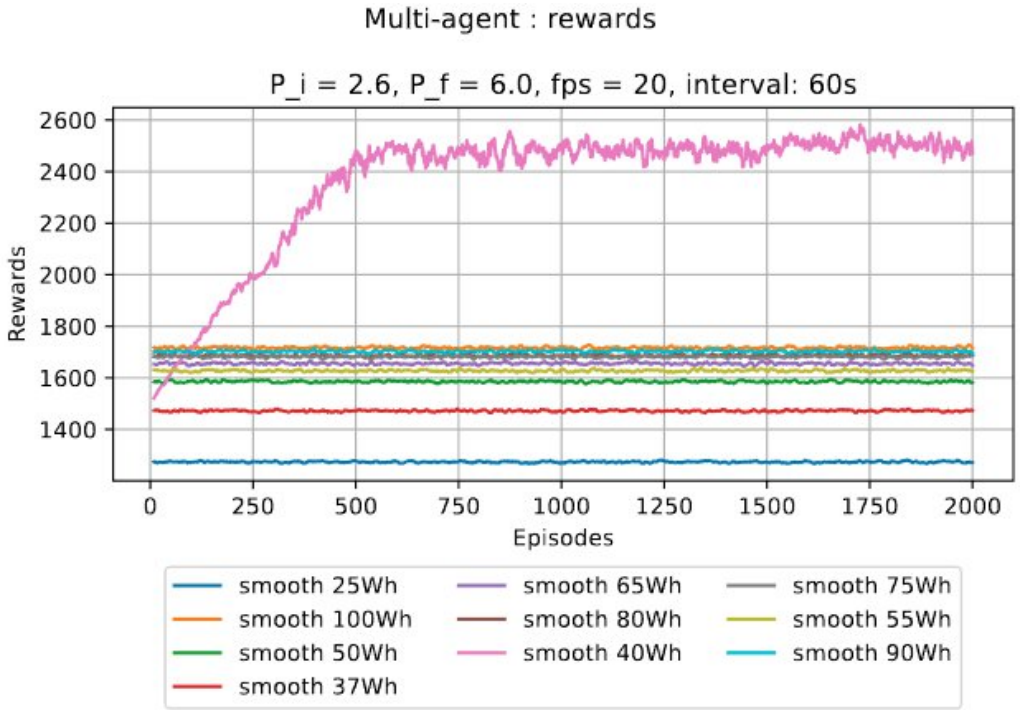




Fourth test: intelligent 40Wh

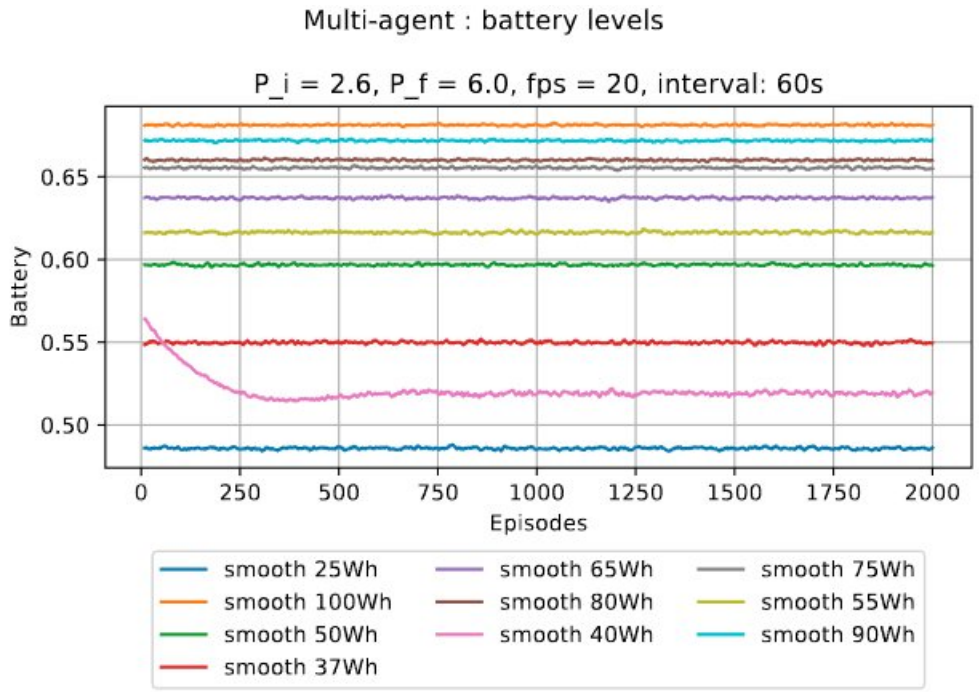
- Imported trained model of node with 37Wh

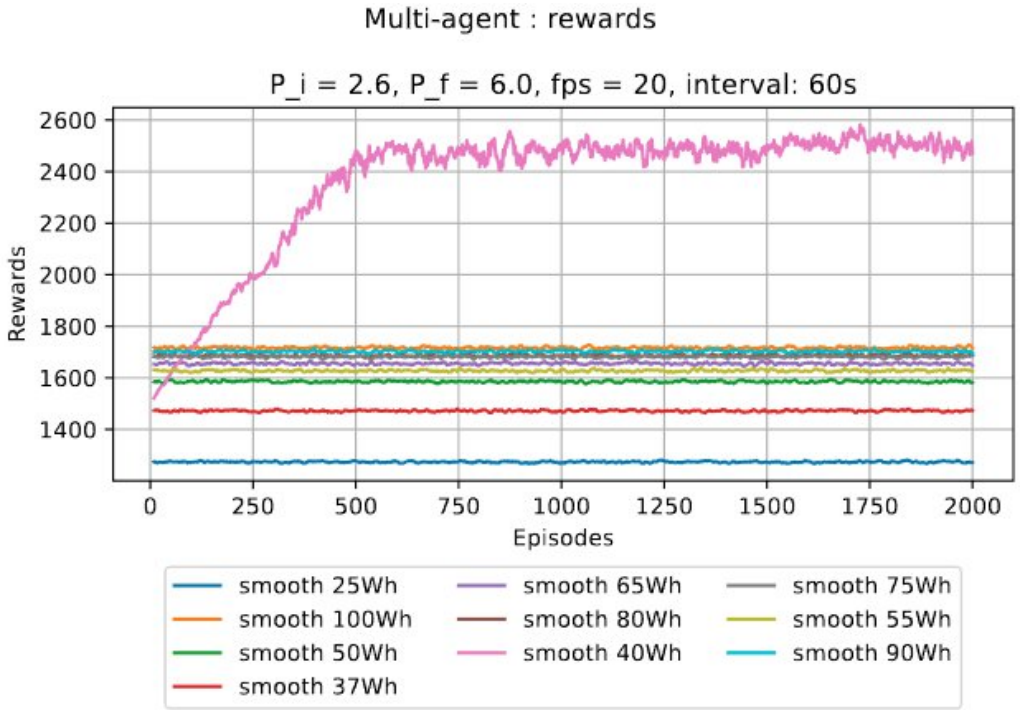




Fourth test: intelligent 40Wh

- Imported trained model of node with 50Wh





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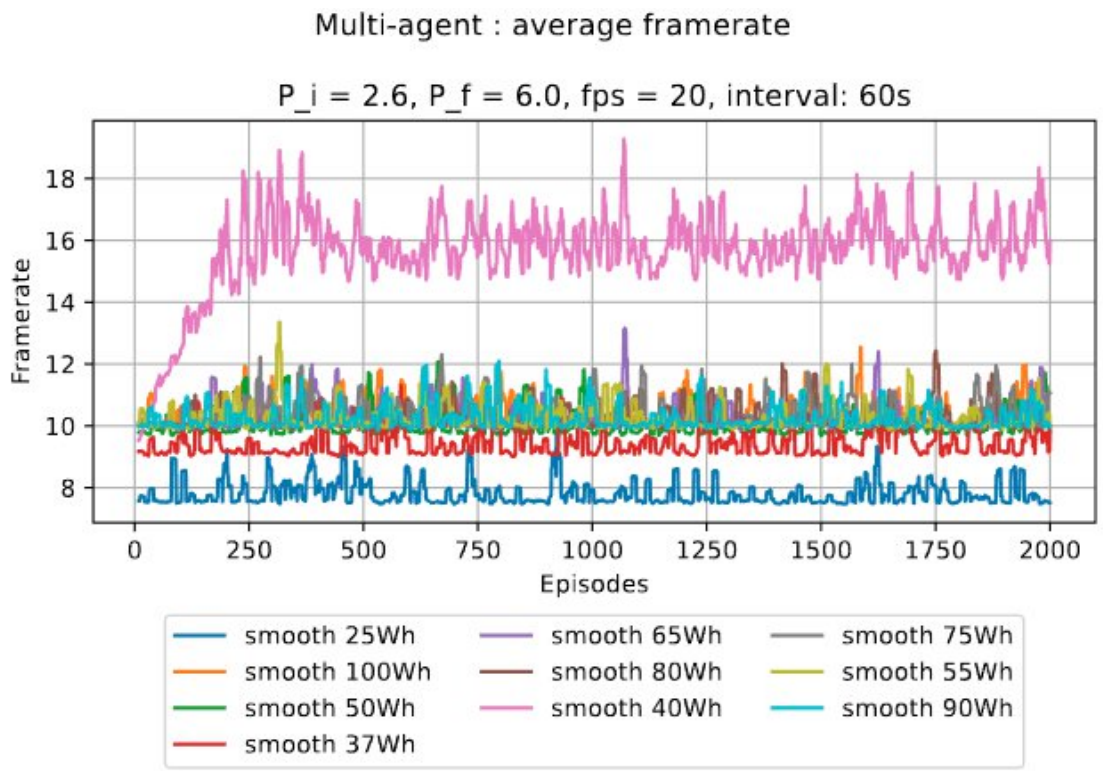
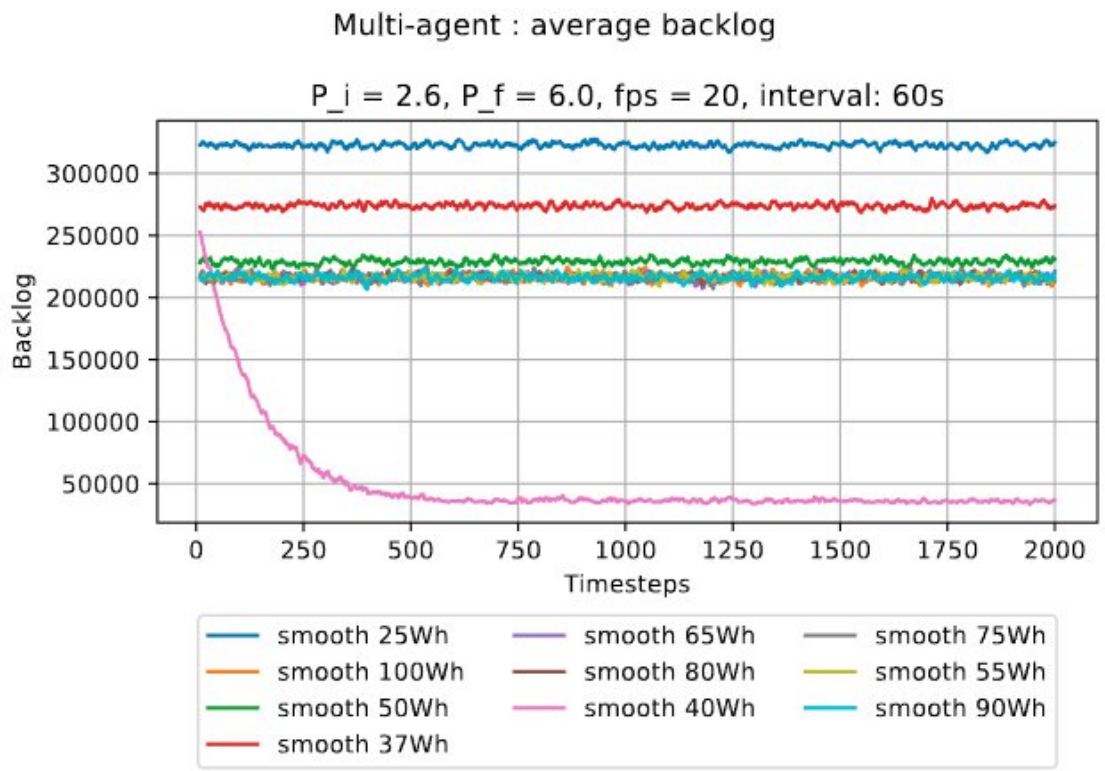
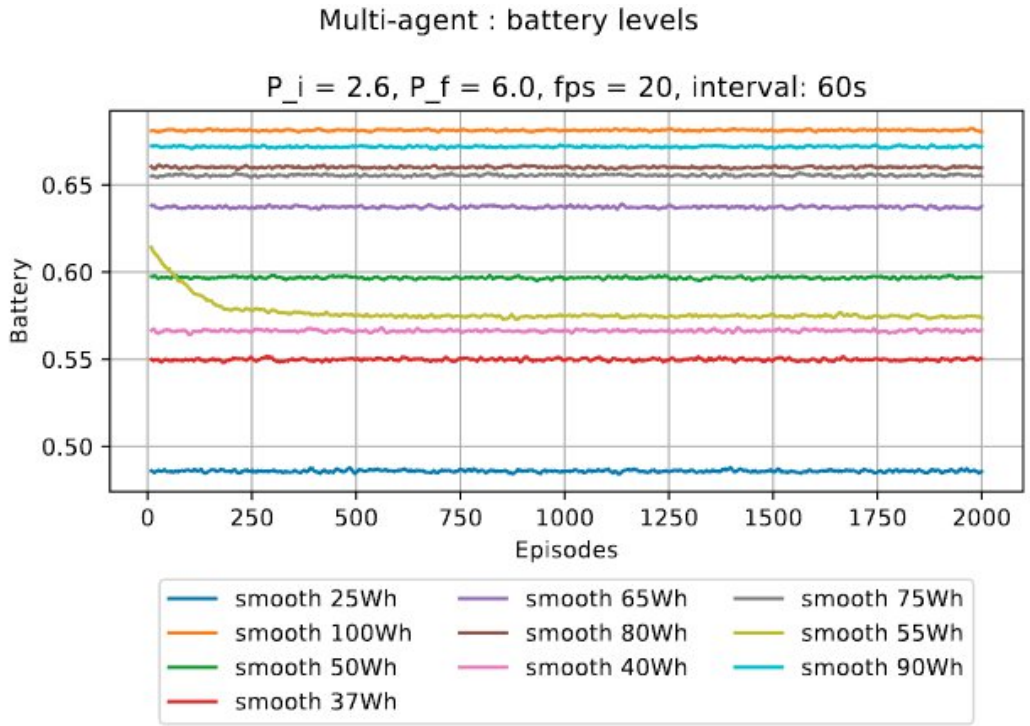
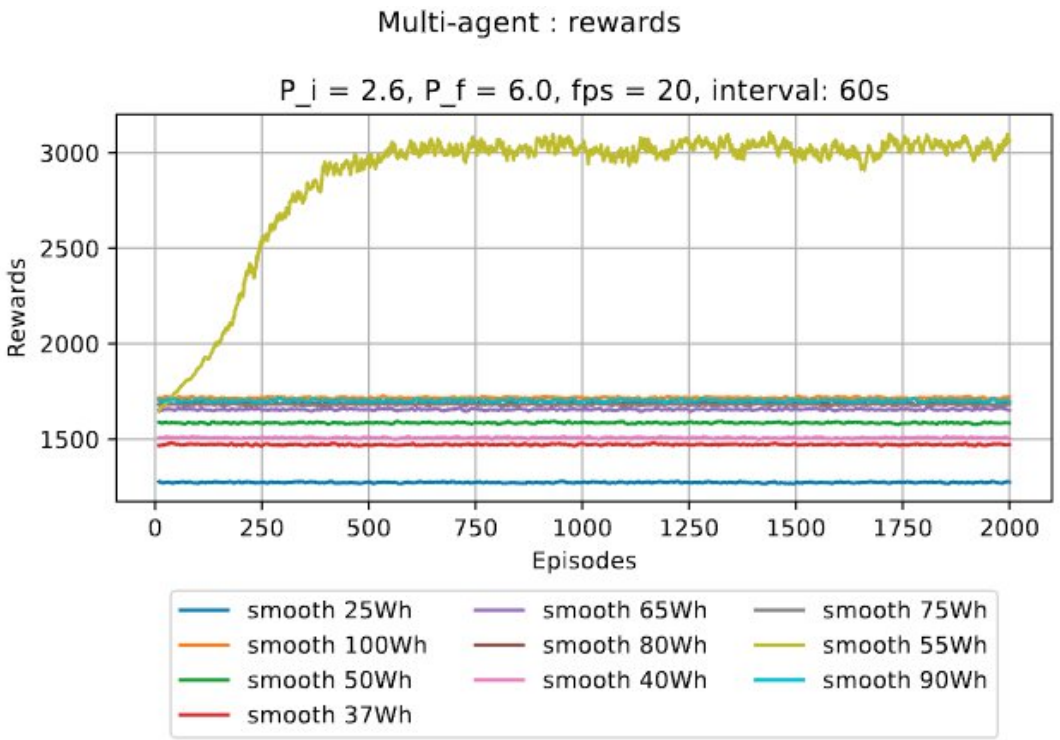




Fourth test: intelligent 55Wh

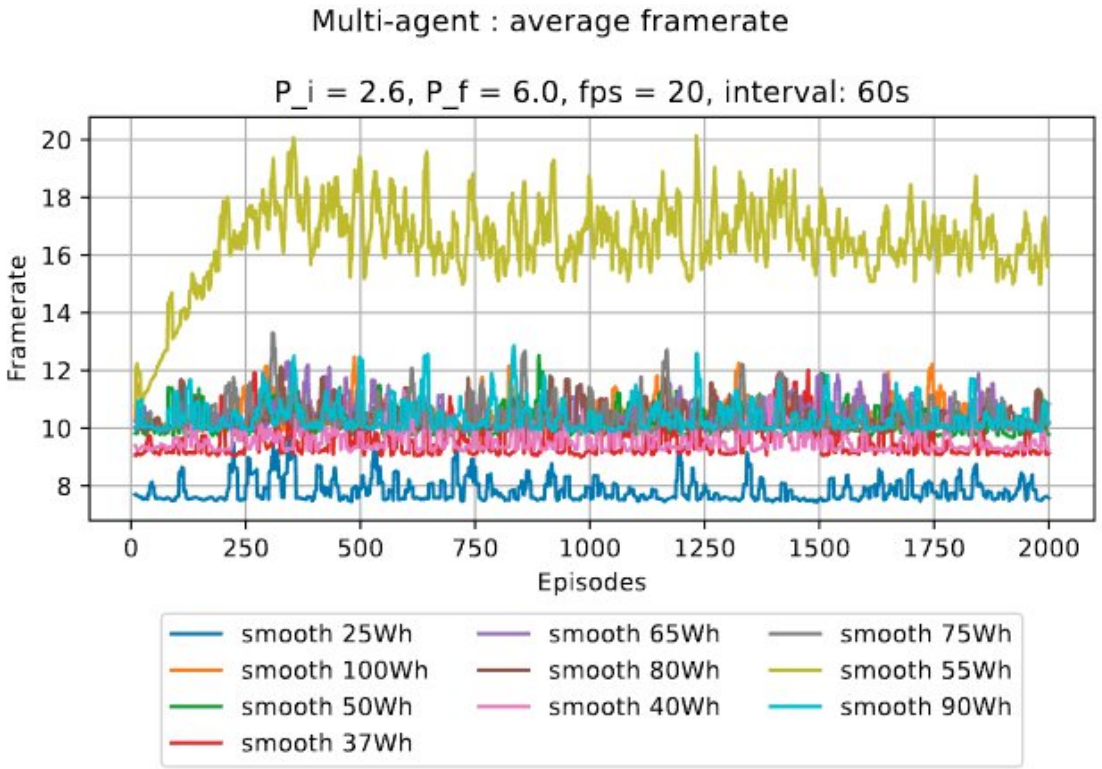
- Imported trained model of node with 40Wh

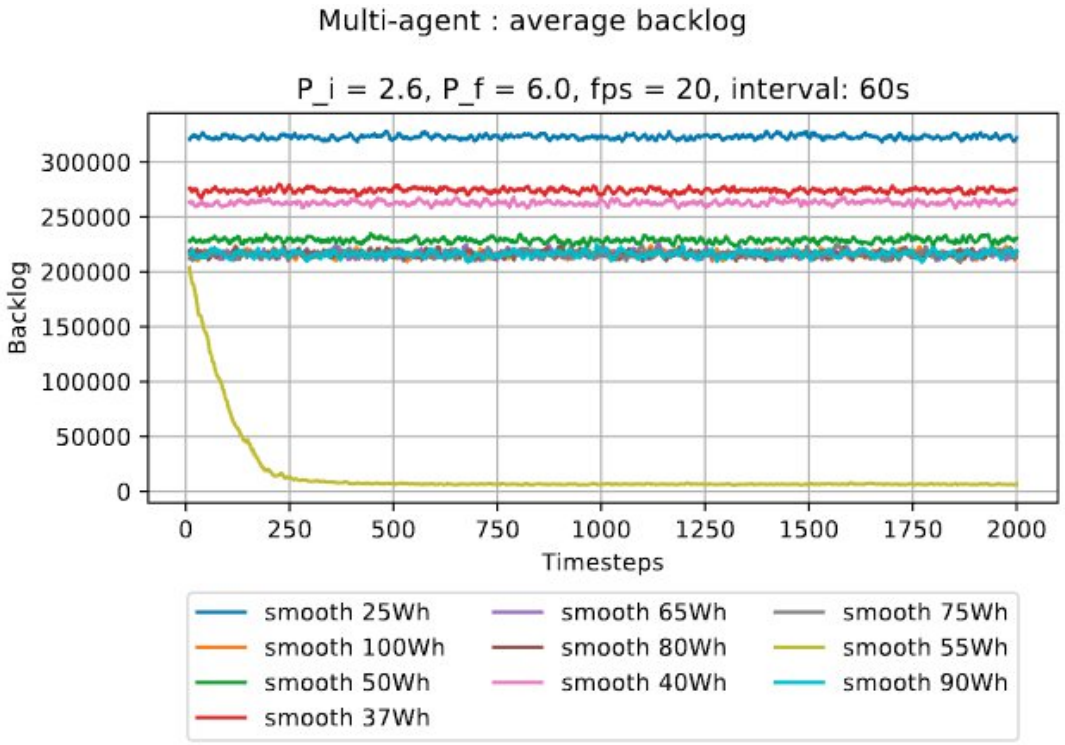




Fifth test: intelligent 55Wh

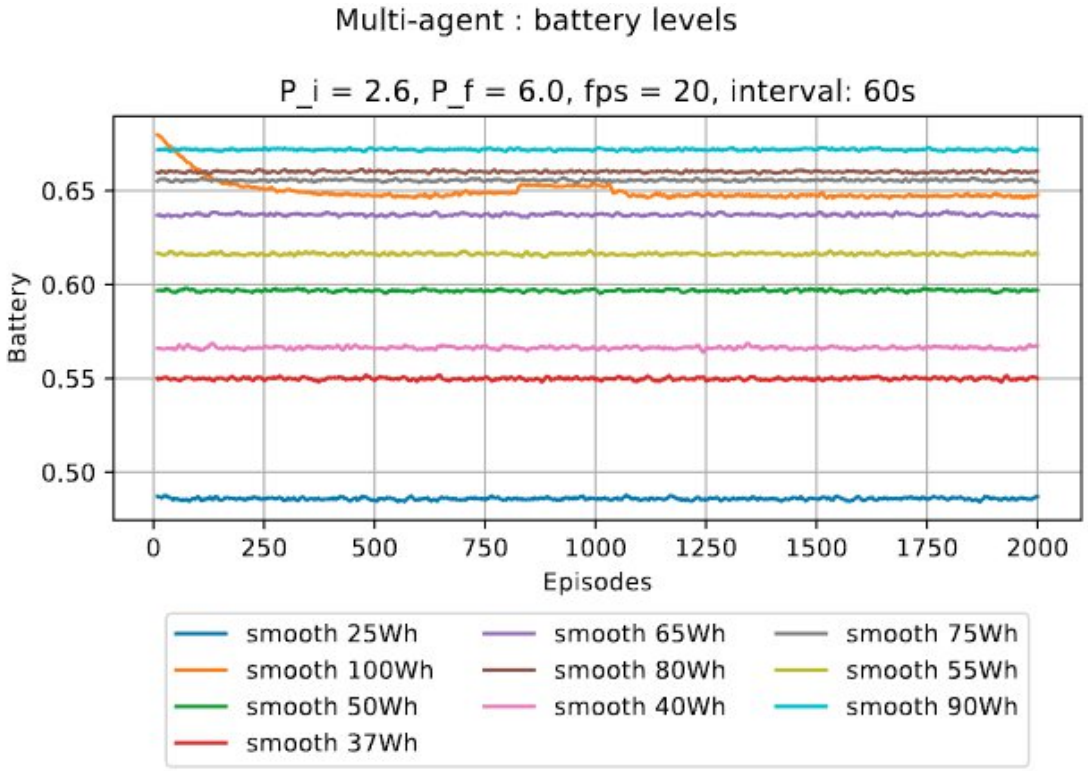
- Imported trained model of node with 40Wh

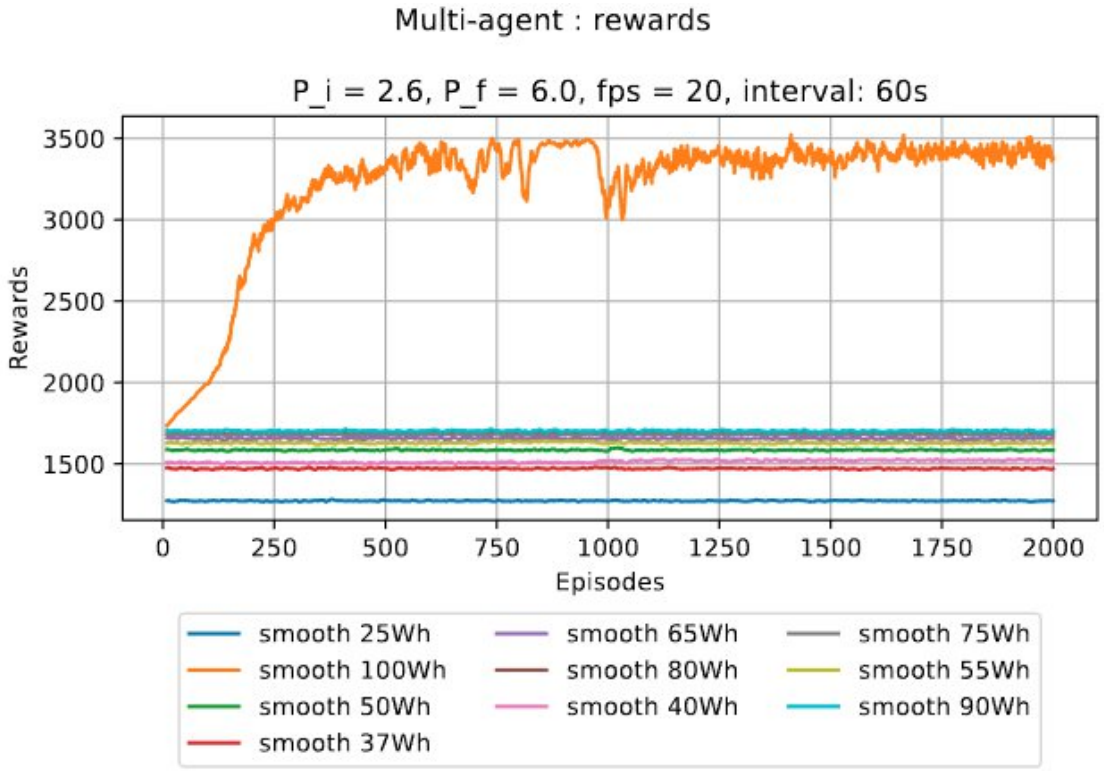




Sixth test: intelligent 100Wh

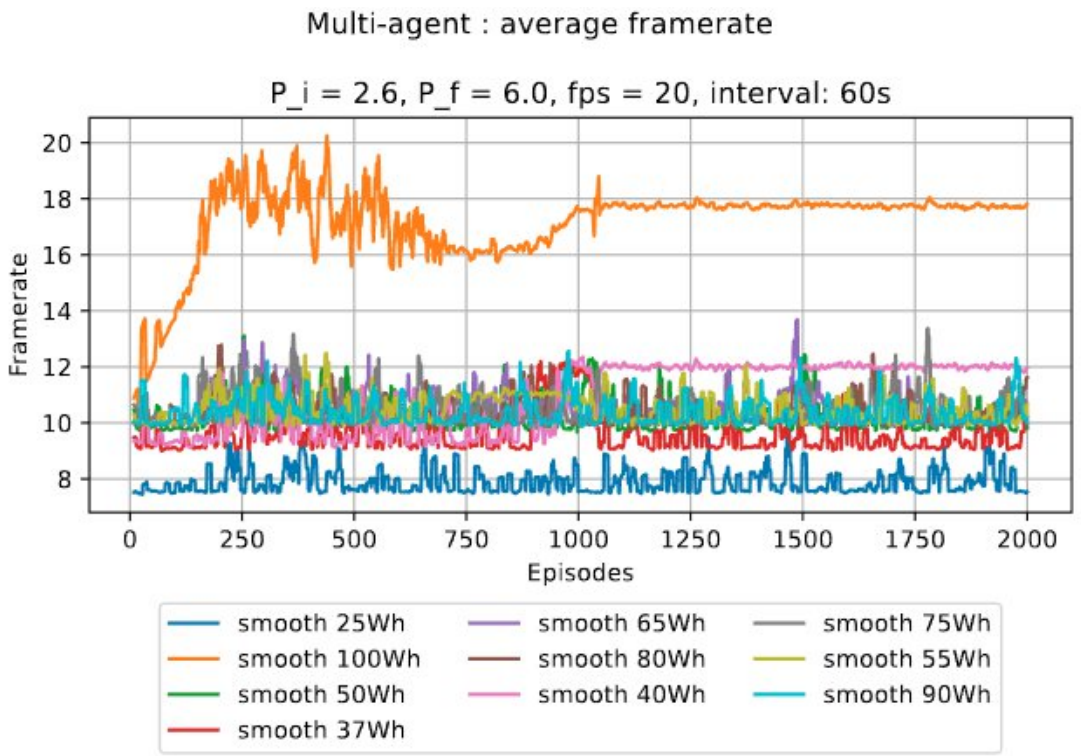
- Imported trained model of node with 55Wh

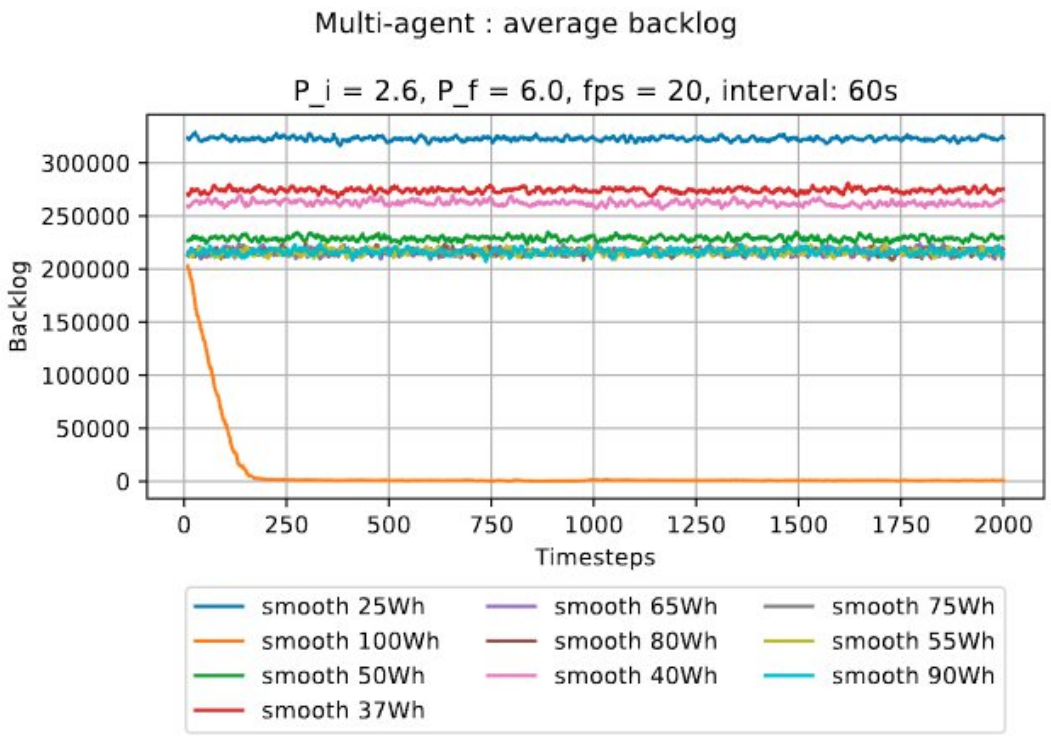




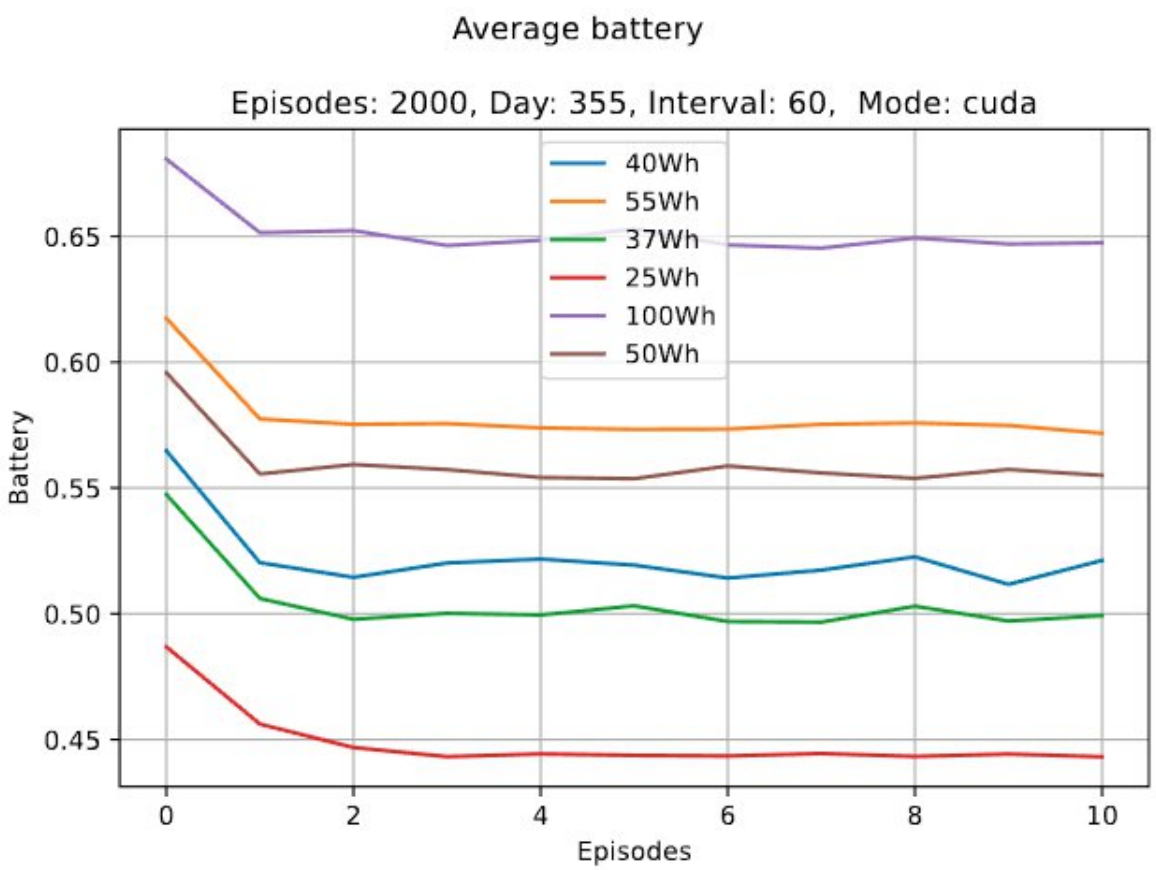
Sixth test: intelligent 100Wh

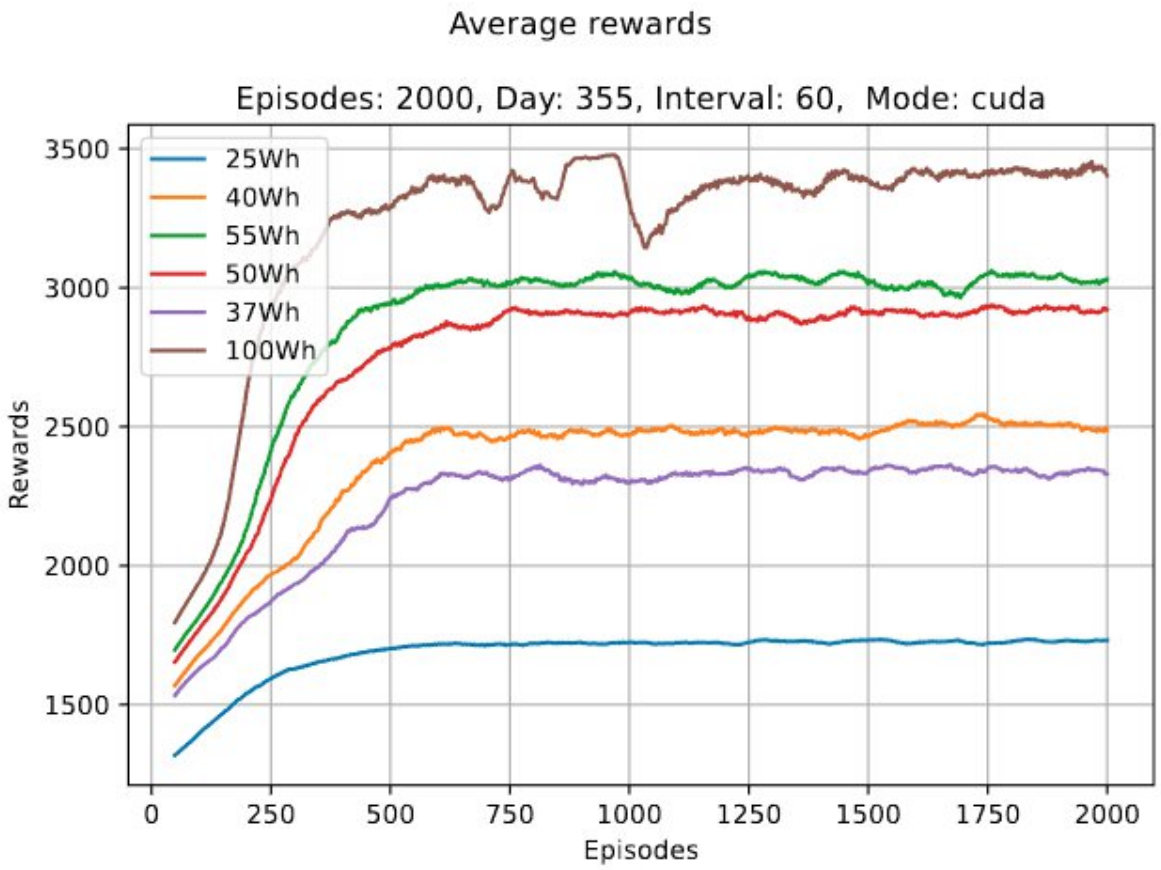
- Imported trained model of node with 55Wh



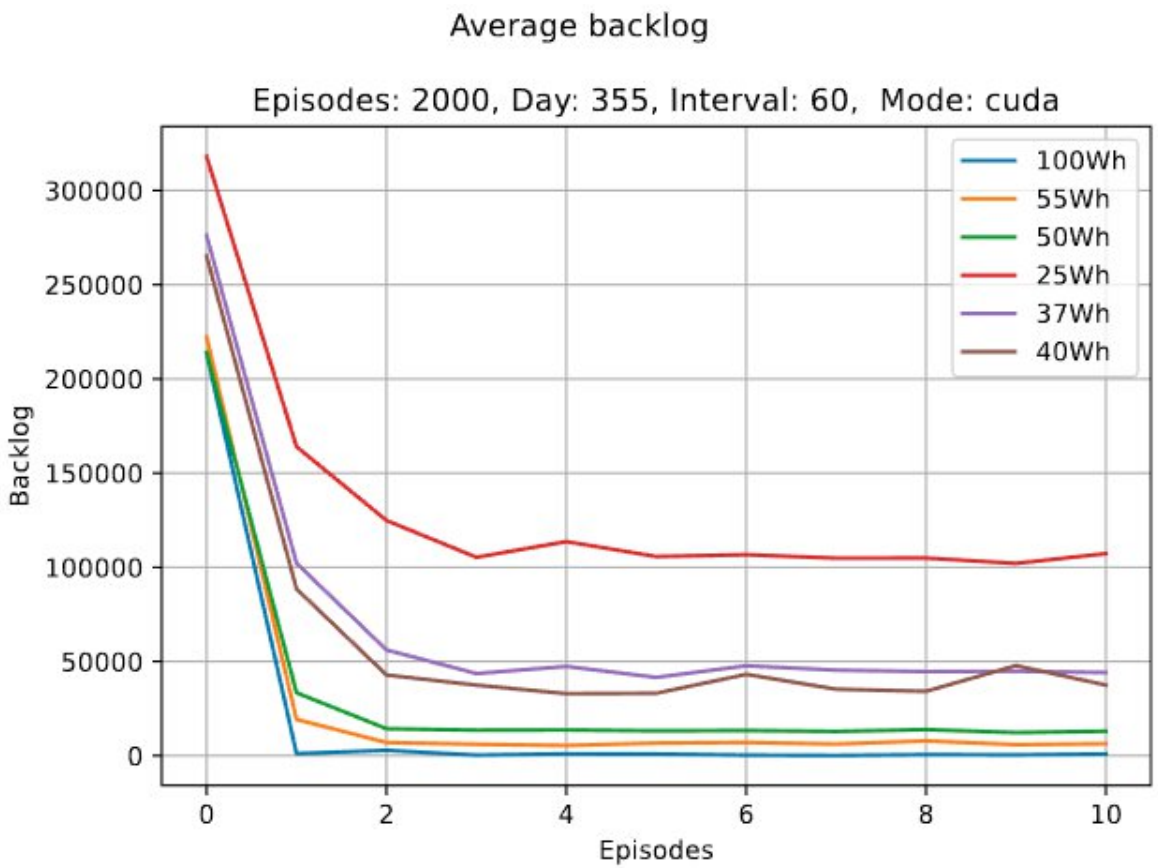


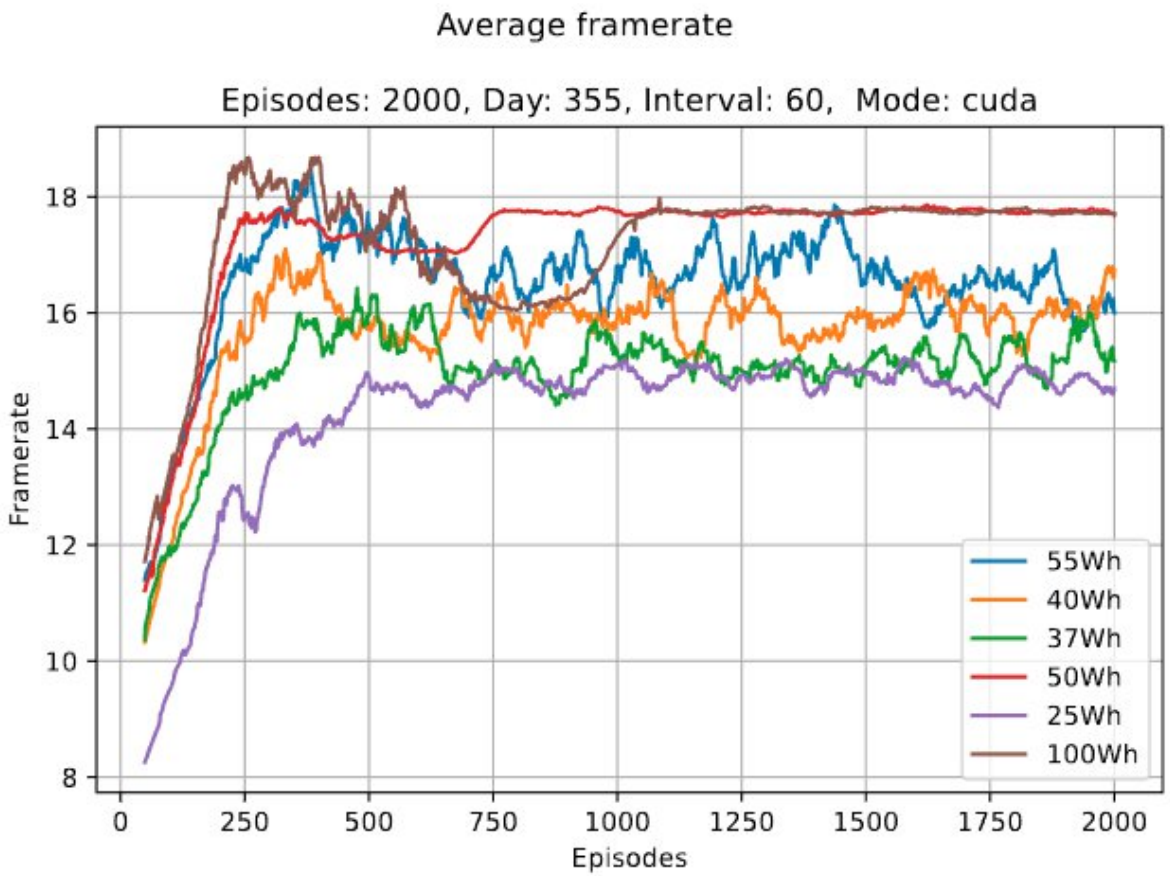
Comparison


Average battery
Episodes: 2000, Day: 355, Interval: 60, Mode: cuda
40Wh
55Wh
37Wh
25Wh
100Wh
50Wh
Battery
0.65
0.60
0.55
0.50
0.45
0
2
4
6
8
10
Episodes


Average rewards
Episodes: 2000, Day: 355, Interval: 60, Mode: cuda
25Wh
40Wh
55Wh
50Wh
37Wh
100Wh
Rewards
3500
3000
2500
2000
1500
0
250
500
750
1000
1250
1500
1750
2000
Episodes

Comparison


Average backlog
Episodes: 2000, Day: 355, Interval: 60, Mode: cuda
Backlog
Episodes
100Wh
55Wh
50Wh
25Wh
37Wh
40Wh


Average framerate
Episodes: 2000, Day: 355, Interval: 60, Mode: cuda
Framerate
Episodes
55Wh
40Wh
37Wh
50Wh
25Wh
100Wh